SIHYUN CHO

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EDUCATION

Kookmin University, Korea *Mar. 2023 – Present*
B.S. student, School of Software, (GPA: 4.14/4.5)

PUBLICATIONS AND MANUSCRIPTS

1. **Sihyun Cho**, Jeongmin Cho, Youngwook Kim, "Pairwise relational perception with object queries for aerial visual grounding", *Manuscript in preparation*, 2026. [Code]
2. **Sihyun Cho**, Changwoo Lee, "Impact of State Design on Actor Comparisons in DRL-Based Task Scheduling for Collaborative Edge Computing", *IAAI 2026* (under review; in Korean). [Code]

RESEARCH EXPERIENCE

Aerial Visual Grounding Research *Jun. 2026 – Present*
Kookmin University, Korea

- Analyzed AerialVG and developed Pairception to inject relational evidence directly into object queries.
- Achieved **SOTA** on AerialVG with 74.3% mean Top-1 accuracy; preparing a journal manuscript.

DRL-Based Task Scheduling Research *Aug. 2026 – Present*
Kookmin University, Korea

- Studied the impact of state design on actor comparisons in DRL-based edge task scheduling.

Summer Research Internship *Jul. 2026 – Aug. 2026*
Kookmin University, Korea — Advisor: Prof. Youngwook Kim

- Extended the capstone research through further model development and experimental analysis.

Federated Learning Research *Sep. 2025 – Jun. 2026*
Kookmin University, Korea

- Conducted research on federated learning and federated low-rank adaptation.

PROJECT EXPERIENCE

Aerial Visual Grounding – Capstone Project *Feb. 2026 – May 2026*
Kookmin University, Korea

- Reviewed visual grounding literature from ViT to Grounding DINO and built an AerialVG model.
- Improved Top-1 accuracy from approximately 50% to 59%.

AWARDS AND HONORS

- **Grand Prize**, AI × Seoul Innovation Content Competition, Sungkyunkwan University, Korea. Nov. 2025

SKILLS

Python, PyTorch, C++